PHYSICS – BOARD QUESTION PAPER : JULY 2023

Time: 3 Hours

Max. Marks: 70

General Instructions:

The question paper is divided into four sections.

1. Section A: Q1 contains Ten multiple choice questions. Q2 contains Eight very short questions.
2. Section B: Q3 to Q14 contain Twelve questions. Attempt any Eight.
3. Section C: Q15 to Q26 contain Twelve questions. Attempt any Eight.
4. Section D: Q27 to Q31 contain Five questions. Attempt any Three.
5. Use of log table is allowed. Use of calculator is not allowed.
6. Figures to the right indicate full marks.
7. For MCQs, it is mandatory to write the correct answer with its alphabet. Only first attempt is valid.

Physical Constants:

$c = 3 \times 10^{8}$ m/s

$h = 6.63 \times 10^{-34}$ Js

$\pi = 3.142$

$g = 9.8$ m/s^2

$\varepsilon_0 = 8.85 \times 10^{-12}$ C^2/Nm^2

$\mu_0 = 4\pi \times 10^{-7}$ Wb/Am

$R_H = 1.097 \times 10^{7}$ m^{-1}

R = 8.31 J/mol·K

SECTION – A

Q.1 Select and write the correct answer: (10M)

(i) A body performing uniform circular motion has constant:

(a) velocity (b) kinetic energy (c) displacement (d) acceleration

(ii) When common salt is dissolved in water, surface tension:

(a) decreases (b) becomes zero (c) remains same (d) increases

(iii) Time period of angular oscillations of bar magnet is proportional to:

(a) I/B (b) B/I (c) $\sqrt{I/B}$ (d) $\sqrt{B/I}$

(iv) Equality of absorption & emission coefficient at same temperature is:

(a) Stefan's law (b) Newton's cooling law

(c) Kirchhoff's law (d) Boyle's law

(v) Second law of thermodynamics deals with transfer of:
(a) work (b) energy (c) pressure (d) heat

(vi) Transverse nature of light is proved by:
(a) Reflection (b) Interference (c) Diffraction (d) Polarization

(vii) A conductor of length L moves with velocity v in magnetic field 2B. Induced emf is:
(a) BLv/4 (b) BLv/2 (c) BLv (d) 2BLv

(viii) Solar cell works on:
(a) diffusion (b) recombination (c) photovoltaic action (d) photoelectric effect

(ix) In SHM, force = 0.2 N at displacement 4 cm. Force constant:
(a) 2 N/m (b) 2.5 N/m (c) 5 N/m (d) 8 N/m

(x) Solenoid of length 25 cm, 250 turns, current 1 A. Magnetic induction =
(a) 0.12568×10^{-3} T (b) 1.2568×10^{-3} T
(c) 1.2568×10^{2} T (d) 1.2568×10^{4} T

Q.2 Answer the following: (8M)
(i) State formula for end correction in resonance tube.
(ii) Classify H_2O and CO_2 into polar and non-polar dielectrics.
(iii) Define potential gradient.
(iv) Find time period for SHM with $v_{max} = 0.08$ m/s, $a_{max} = 0.32$ m/s^2.
(v) Define gyromagnetic ratio.
(vi) State conditions for current and impedance in parallel resonance.
(vii) Radius of third Bohr orbit is 0.477 nm. Find radius of second orbit.
(viii) Name logic gate having single input–single output.

SECTION – B

Attempt any Eight questions. (16M)
Q.3. State conditions for steady interference pattern.
Q.4. Derive expression for electric field due to infinitely long straight charged wire.
Q.5. What are eddy currents? Write two applications.
Q.6. Define and state formulae for inductive and capacitive reactance.
Q.7. Draw p–V diagram. Explain positive and negative work.
Q.8. State law of length and law of linear density for vibrating string.
Q.9. For moving coil galvanometer, show that S = G/(n−1).

Q.10. Calculate ratio of kinetic energies if electron and proton have same de-Broglie wavelength.

Q.11. Derive expression for half-life using law of radioactive decay.

Q.12. Motorcyclist rides inside well of death of radius 4 m, $\mu = 0.4$. Calculate maximum period.

Q.13. Diameter of water drop for excess pressure 80 N/m² ($\sigma = 7.2\times10^{-2}$ N/m).

Q.14. Magnetic field 16 μT at 2.4 cm from long straight wire. Find current.

SECTION – C

Attempt any Eight questions. (24M)

Q.15. Explain construction and working of Ferry's black body.

Q.16. Show deflection in galvanometer ∝ current.

Q.17. Explain half-wave rectifier with circuit and waveforms.

Q.18. Obtain expression for time period of simple pendulum.

Q.19. Show beat frequency = $|f_1 - f_2|$.

Q.20. Obtain expression for orbital magnetic moment of electron.

Q.21. State Einstein's photoelectric equation and explain any two characteristics.

Q.22. Calculate shortest Paschen wavelength and longest Balmer wavelength.

Q.23. A 60W filament lamp radiates energy. Emissivity = 0.5, area = 5×10^{-5} m². Find temperature.

Q.24. Two capacitors C_1 and C_2 in parallel, in series with C_3. Find equivalent capacitance.

Q.25. Meter-bridge: X in left gap, 30Ω right gap, null at 40 cm. Find X and new null point if both resistances increase by 15Ω.

Q.26. Inductor 200 mH connected to AC source (peak emf 210 V, f = 50 Hz). Find peak current and instantaneous voltage when current is maximum.

SECTION – D

Attempt any Three questions. (12M)

Q.27. State and prove theorem of parallel axes.

Q.28. Define isothermal and adiabatic process. One mole gas at 1 mPa, 27°C expands isobarically to double volume. Calculate work done.

Q.29. Distinguish streamline flow and turbulent flow. Calculate terminal velocity of 0.4 mm bubble rising through liquid with viscosity 0.1 Ns/m², density 900 kg/m³ (air density 1.29 kg/m³).

Q.30. What are Fraunhofer and Fresnel diffraction? Plane wave λ = 5500Å. Distance of 10 bright fringes = 2 cm on screen 2 m away. Find slit separation.

Q.31. What is transformer? State working principle and distinguish between step-up and step-down transformers.